Ecosystem, Heuristics & Industry Mapping

How `salesforce-agentforce-enterprise-bridge` maps to the Modern AI Ecosystem, the 20 AI Engineering Heuristics, and real industry use-cases

> **Important**
>
> **Honesty legend used throughout** — ✅ **Implemented** (in the repo, tested) · 🟡 **Partial** (simulated or minimal version) · 🚧 **Roadmap** (designed seam, not built) · ⛔ **N/A (hosted)** (provider-side concern unless you self-host models).
>
> The reference implementation is deliberately narrow: it is a **governed action plane**, not a full AI platform. Claiming more would not survive a 2:1 panel.

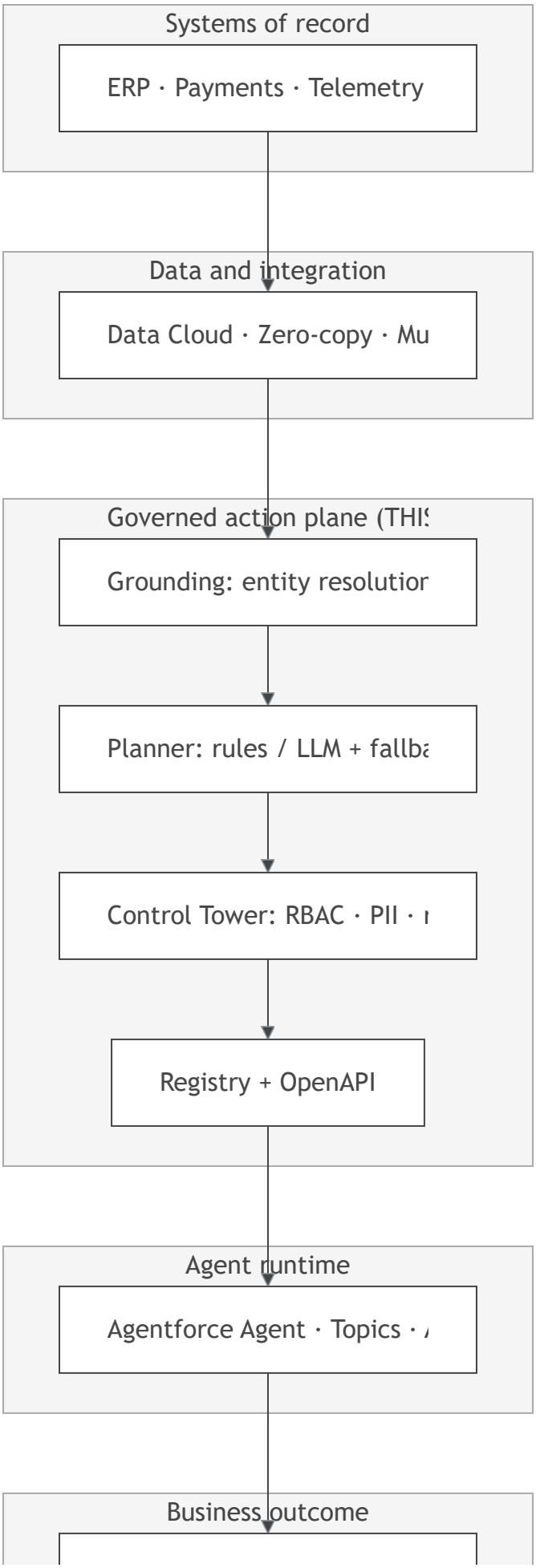
1 · The one-paragraph answer

Most AI roadmaps list *tools*. The bridge targets the part that decides whether AI ever reaches production in a regulated enterprise: **turning an event into a trustworthy decision and a governed action, with proof.**



It touches **7 of the 9 ecosystem areas** (LLM, agentic, RAG, embeddings, security, observability, memory — at differing depths), ships **2 of 20 heuristics fully** (tool use, structured output), **5 partially**, and has explicit seams

for **8 more**; **5 are serving-layer techniques** that only matter if you self-host models.


Systems of record
ERP · Payments · Telemetry
Data and integration
Data Cloud · Zero-copy · Mu
Governed action plane (THIS
Grounding: entity resolutior
Planner: rules / LLM + fallba
Control Tower: RBAC · PII · r
Registry + OpenAPI
Agent runtime
Agentforce Agent · Topics · A
Business outcome

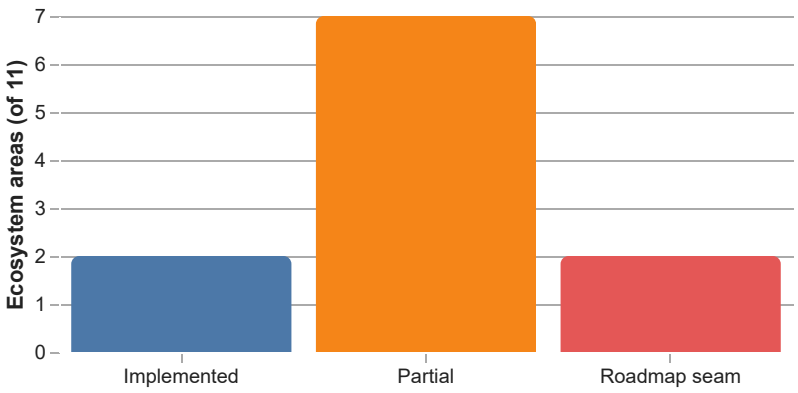
Hours saved · risk avoided ·

2 · Modern AI Ecosystem — coverage map

The ecosystem graphic has **11 branches** (nine in the text post plus *Vector Database* and *Automation* shown in the image).

#	Ecosystem area	Status	What the solution does	Where	Production swap / next step
1	**Foundation models (LLMs)**	✅	Claude via Messages API; provider-neutral `Planner` protocol	`agent.py::AnthropicPlanner`	Any provider/Ollama/vLLM; Salesforce Models API to stay inside one trust boundary
2	**Agentic AI**	🟡	Planner ⊥ tools separation; Agentforce is the agent runtime; deterministic DAG for one decision	`pipeline.py`, `actions.py`	LangGraph/Strands for multi-step tool loops (same Control Tower)
3	**RAG**	🟡	Account-scoped retrieval of similar incidents feeding the planner	`grounding.py`	Data Cloud search index / RAGFlow / GraphRAG for relationship-heavy data
4	**Embedding models**	🟡	Deterministic feature-hashing embeddings (512-d, signed, L2-norm)	`grounding.py::embed`	OpenAI/Cohere/Voyage/BGE/Vertex embeddings; Data Cloud vectorisation
5	**MCP**	🚧	Action manifest (name, JSON-Schema, risk, permission) is *MCP-tool-shaped*	`/v1/actions`, `/openapi.json`	Expose the registry as an MCP server; consume Salesforce-hosted MCP servers
6	**AI security**	✅/🚧	PII redaction (SSN, Luhn card, email, phone), prompt-injection containment, RBAC, allow-listed actions, SOQL escaping	`pii.py`, `control_tower.py`, `rbac.py`	Presidio (NER), NeMo Guardrails / Guardrails AI, Lakera Guard, Bedrock/Azure content safety
7	**Observability**	🟡	Decision counters, latency quantiles, fallback & mirror-failure counters, readiness, audit trail	`metrics.py`, `audit.py`	Langfuse / Phoenix / LangSmith traces; Ragas/TruLens for RAG quality; OpenTelemetry

#	Ecosystem area	Status	What the solution does	Where	Production swap / next step
8	**Memory**	🟡	Bounded per-account history (short-term, entity-scoped); audit as durable record	`grounding.py`	Data Cloud profile as long-term memory; Mem0/Zep/Letta for conversational memory
9	**Agent frameworks**	🟡	Typed contracts with pydantic (PydanticAI-style discipline); Agentforce as managed framework	`models.py`	OpenAI Agents SDK / Semantic Kernel / Google ADK / Bedrock Agents where the customer standardises
10	**Vector database**	🟡	In-memory bounded index per account	`VectorIndex`	pgvector / Data Cloud / Pinecone / Qdrant / Elasticsearch
11	**Automation / orchestration**	🔀	Async bus, retry, breaker, DLQ	`bus.py`, `resilience.py`	Temporal / Airflow / Prefect for durable retries & replays; n8n / Power Automate for citizen-built flows; Kafka for ingest



> **Note**
>
> **Senior-engineer takeaway from the roadmap graphic:** *"Don't just learn AI tools; learn how to connect them to solve real-world problems."* The bridge is exactly that connective tissue — contracts, grounding, governance, and audit between the model and the business system.

3 · The 20 AI engineering heuristics — applied

Each heuristic is a *pressure → technique* pair. Below: whether this solution already applies it, how, and where it would bite as you scale.

3.1 Context & retrieval (1–4)

#	Pressure → Technique	Status	In this solution
1	New knowledge + generation → **RAG**	🟡	Planner receives *retrieved, account-scoped* facts (event count, exposure, similar incidents ≥ 0.5 cosine) instead of free recall
2	Keyword + semantic → **Hybrid search**	🟡	Entity resolution is exact + fuzzy (lexical); similarity is hashed-lexical. Add BM25 + dense embeddings with rank fusion
3	Retrieval + noise → **Reranker**	🏜️	Today: top-k with min score. Add a cross-encoder reranker once the corpus is large (retrieve 50 → rerank to 5)
4	Long prefix + many calls → **Prompt cache**	🏜️	System prompt is static by design (cache-friendly). Enable provider prompt caching; keep variable event JSON *after* the stable prefix

3.2 Training & adaptation (5–8)

#	Pressure → Technique	Status	In this solution
5	Labels + behaviour → **SFT**	🏜️	Every human **approve/reject/override** is a labelled (event, context)→action example — free supervision
6	Fine-tune + single GPU → **LoRA**	🏜️	If a small open model replaces the API planner, adapt with LoRA on the override-derived dataset
7	Verifiable output + no labels → **GRPO**	🏜️	Rewards are auto-checkable: *schema-valid* ∧ *action-allowed* ∧ *not-overridden* ∧ *idempotent*. A natural RL-with-verifiable-rewards setup
8	Big model + cheap inference → **Distillation**	🏜️	Distil frontier-model decisions into a small, cheap triage model; keep the large one for hard cases

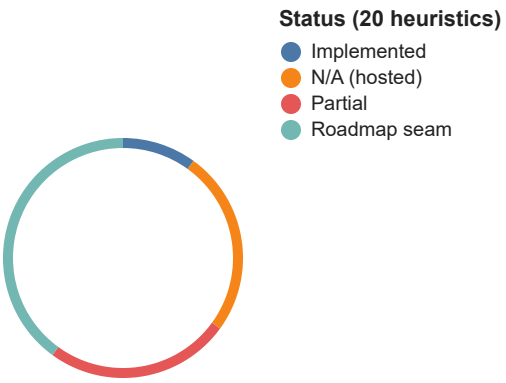
3.3 Inference & serving (9–16)

#	Pressure → Technique	Status	In this solution
9	Big model + OOM → **Quantization**	⛔	Hosted API; relevant only if self-hosting (e.g., vLLM/Ollama on-prem for data-sovereignty customers)
10	Long generation + latency → **KV cache**	⛔	Provider-side. Outputs here are short JSON (≤ 512 tokens), so decode cost is small
11	Memory + fragmented → **PagedAttention**	⛔	vLLM feature; n/a to hosted usage
12	Draft + target → **Speculative decoding**	⛔	Provider/server feature; n/a
13	Variable requests + throughput → **Continuous batching**	🟡	*App-layer analogue:* bounded queue + worker pool keeps workers busy under variable load, with 429 backpressure
14	Small + large model → **Model routing**	🟡	Rules-first baseline, LLM as upgrade, automatic fallback. Next: route *only low-confidence* events to the LLM
15	Meaning + cache → **Semantic cache**	🏞️	Recurring alerts ("gateway outage…") are highly repetitive → cache decisions by embedding similarity with TTL and per-account scope
16	Long prompts + fast tokens → **Disaggregated serving**	⛔	Self-hosting concern (prefill vs decode pools); n/a

3.4 Agents & quality (17–20)

#	Pressure → Technique	Status	In this solution
17	Model + external APIs → **Tool use**	✅	`ActionRegistry`: typed, permissioned, risk-tiered tools exposed via OpenAPI; model can only *request*, Control Tower *executes*
18	Context + overflow → **Sub-agents**	🏞️	Pipeline is single-step; per-account scoping is the analogue. For multi-step investigations delegate to sub-agents that return *summaries*, not raw logs

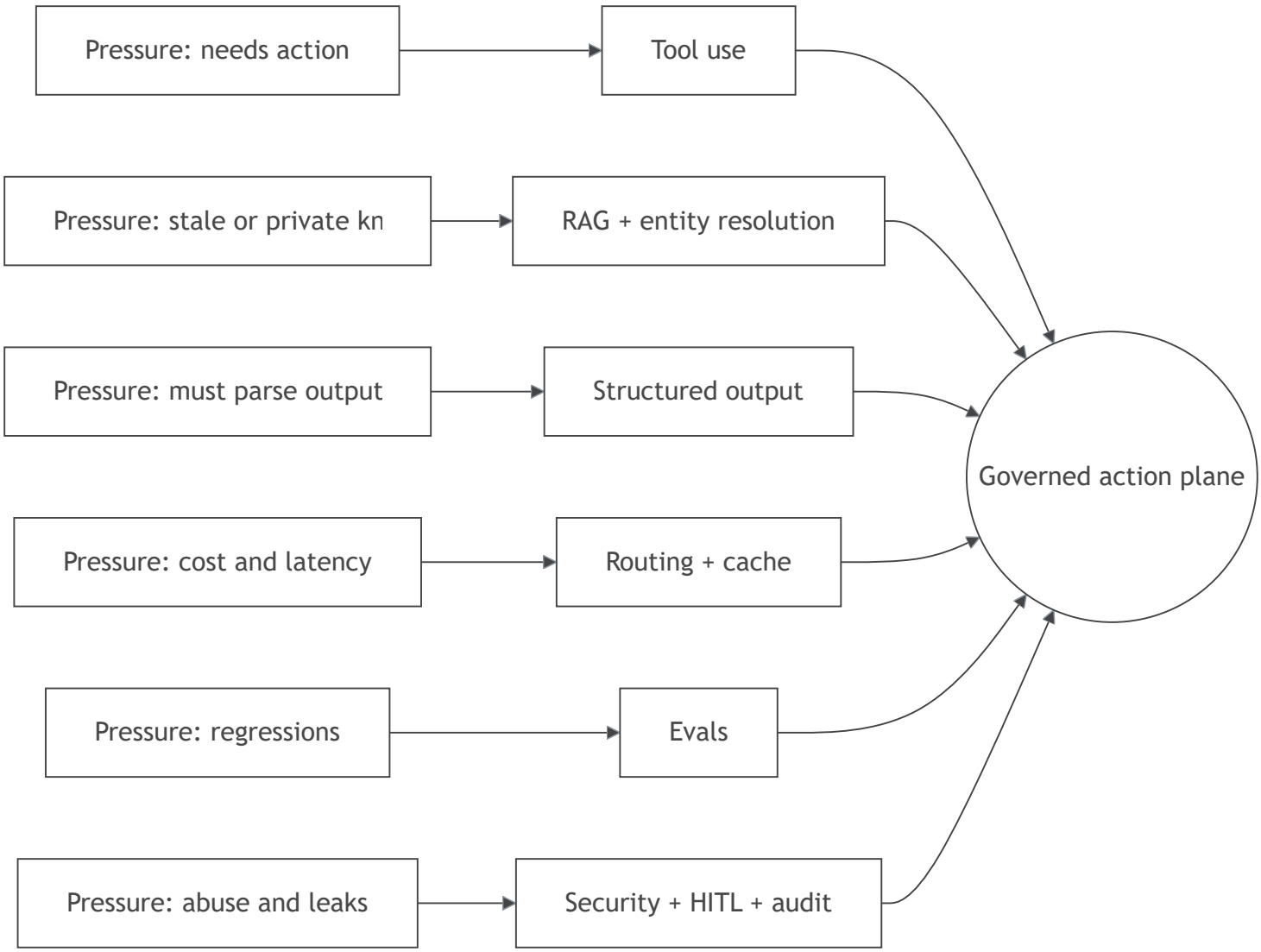
#	Pressure → Technique	Status	In this solution
19	Text + JSON → **Structured output**	✅	JSON-only planner contract → pydantic `Plan`; params re-validated against action schemas; invalid → fallback/reject
20	Prompts + regression → **Evals**	🟡	47 deterministic tests incl. adversarial planner outputs (100 % line+branch). Golden-set evals + shadow mode are the next gate



4 · “Name the pressure” — five tensions this solution resolves

> *“Once you can name the pressure, the technique becomes obvious.” — use this framing in the interview.*

Production pressure	Naive failure	Technique(s) applied here	Evidence
Model knows things, can’t act safely	Free-form tool calls	Tool use + structured output + governance	`ActionRegistry`, `ControlTower`
Fluent but ungrounded	Hallucinated customer facts	RAG-style grounding, entity resolution, tenant-scoped retrieval	`GroundingService`
Non-determinism vs. audit	“Why did it do that?”	Deterministic baseline, full audit, schema-validated proposals	`RuleBasedPlanner`, `AuditLog`
Cost/latency vs. quality	LLM on every event	Rules-first routing, fallback, (roadmap) semantic cache + routing	`ResilientPlanner`
Prompt changes break behaviour	Silent regressions	Evals-as-tests, adversarial outputs, 100 % coverage gate; (roadmap) golden set	`tests/`, CI


Pressure: needs action
Tool use
Pressure: stale or private kn
RAG + entity resolution
Pressure: must parse output
Structured output
Governed action plane
Pressure: cost and latency
Routing + cache
Pressure: regressions
Evals
Pressure: abuse and leaks
Security + HITL + audit

5 · The real-world problem

Every industry has the same shape of problem:

> ***High-volume operational signals arrive from many systems; a small fraction need an action in a system of record; some actions are consequential; a regulator or auditor will ask who decided and why.***

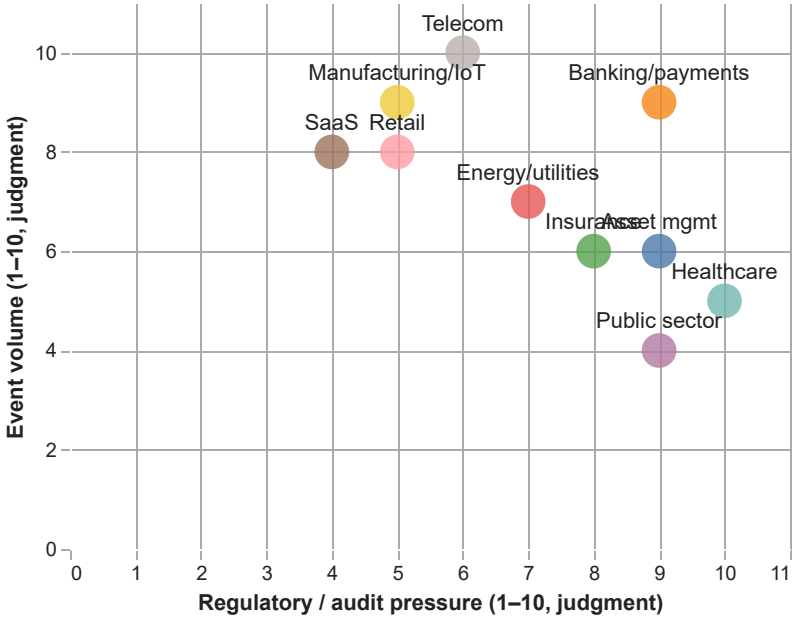
Pain today	Consequence	What the bridge changes
Signals fragmented across ERP / payments / IoT / logs	Slow, inconsistent triage; duplicate work	Contract + dedupe + entity resolution → one unified view per customer
AI pilots stall at security/compliance review	Value never reaches production	PII stripping, RBAC, risk tiers, human approval, audit — *built in, testable*
Agents act on stale/untrusted context	Wrong or unsafe actions	Account-scoped grounding; validated, allow-listed actions
No evidence trail	Audit findings; regulatory exposure	`Bridge_Audit_Log__c` for every decision (executed, pending, denied, rejected, failed)
Integration brittleness	Outages cascade	Backoff + jitter, circuit breaker, DLQ, outbox replay

6 · Industry segment mapping

Segment	Operational signal (event)	LOW-risk autonomous action	HIGH-risk (human-approved) action	Salesforce surface	Typical compliance drivers*	Value metric
Asset / wealth management	Trade-break, reconciliation mismatch, KYC/AML alert	Open ops Case, enrich with account context	Financial review hold; escalation to compliance	Financial Services Cloud, Service Cloud	SEC/FINRA recordkeeping, SOX	Break-resolution time; audit-prep hours
Banking & payments	Gateway outage, card-decline spikes, chargeback signals	Create incident Case; notify relationship manager	Merchant account freeze / credit	FSC, Service Cloud	PCI-DSS, GLBA, SOX	Auth-rate recovery; dispute cycle time
Insurance	FNOL/claims anomalies, policy-admin errors, fraud flags	Create claim-exception Case	Payout hold/release; SIU referral	Financial Services Cloud (Insurance)	State insurance regulation, SOX	Claim cycle time; leakage reduction
Healthcare & life sciences	Device telemetry, billing denials, supply alerts	Care-team or billing-ops Case	Patient-account adjustments (PHI-safe)	Health Cloud, Service Cloud	HIPAA, GxP	Denial turnaround; PHI-incident risk avoided
Telecom & media	Network/KPI degradations, provisioning failures	Customer-impact Case, proactive notification	SLA credits / bulk remediation	Communications Cloud	CPNI/privacy, SLA contracts	MTTR; credits avoided
Manufacturing & IoT	Machine telemetry, quality drift, warranty claims	Field-service Case	Line stoppage / recall escalation	Manufacturing Cloud, Field Service	ISO 9001, safety regs	Downtime hours; warranty cost

Segment	Operational signal (event)	LOW-risk autonomous action	HIGH-risk (human-approved) action	Salesforce surface	Typical compliance drivers*	Value metric
Retail & e-commerce	Payment failures, fulfilment exceptions, fraud signals	Order-exception Case	Refund above threshold; account block	Commerce / Service Cloud	PCI-DSS, consumer-protection laws	Cart-recovery; refund leakage
Energy & utilities	Outage/smart-meter anomalies, billing disputes	Service Case, crew-dispatch request	Large-customer billing adjustment	Energy & Utilities Cloud	Utility-commission rules, NERC CIP (ops tech)	Outage-case handling time
Public sector	Benefit-processing exceptions, case backlogs	Triage and route Case	Eligibility/benefit overrides	Public Sector Solutions	FedRAMP-style controls, records laws	Backlog burn-down; auditability
SaaS / high-tech	Product telemetry, churn/health signals, incident feeds	Support/CS Case; success-plan nudge	Credits, contract exceptions	Service / Sales Cloud	SOC 2, GDPR	Time-to-resolve; retention

*Illustrative regimes, not legal advice; confirm per customer.

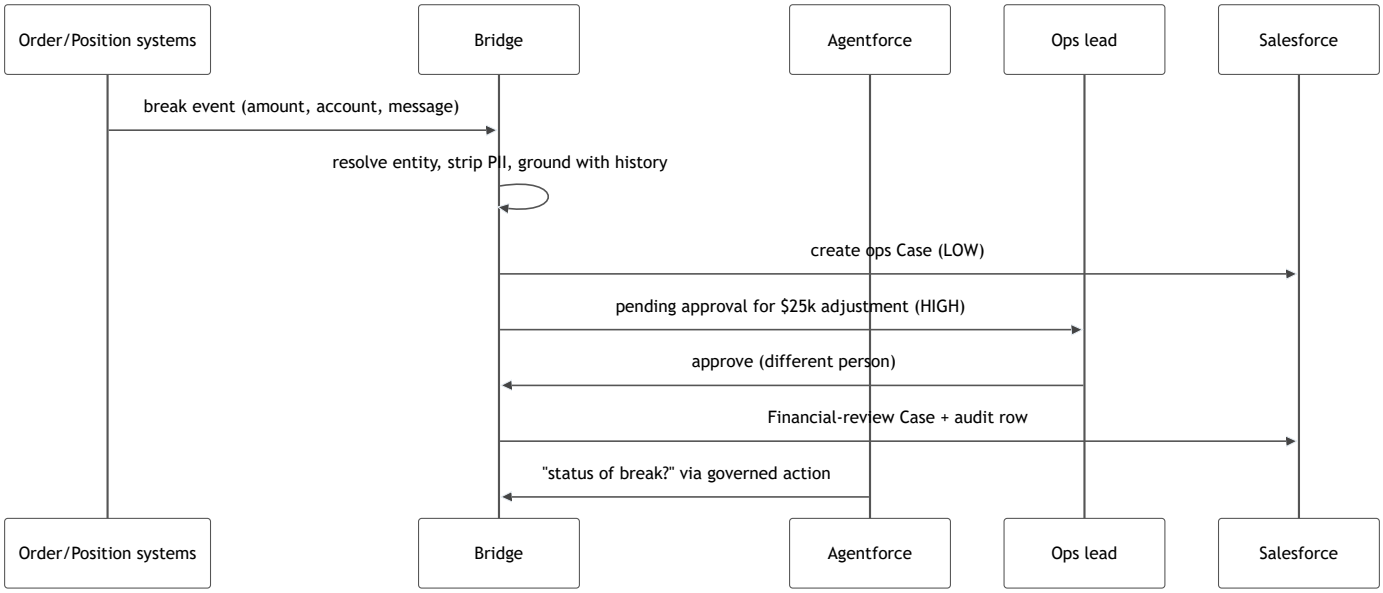


Tip

Sweet spot = top-right (high regulation *and* high volume): banking/payments and asset management — which is also where the candidate's résumé is strongest (J.P. Morgan AM / Highbridge, Balyasny, Millburn). Lead with that; show breadth second.

7 · Worked scenarios (use one in the interview)

7.1 Asset management — reconciliation break (resume-aligned)



Heuristics in play: RAG, tool use, structured output, evals, (roadmap) semantic cache for repeating breaks.

7.2 Banking/payments — gateway outage with card data in logs

Events contain card numbers and emails → **redaction before prompts and CRM**; repeated declines (≥3 events) escalate priority; refunds/freeze actions are HIGH-risk. *Heuristics:* security + HITL, hybrid search over incident text, prompt cache for the static policy prefix.

7.3 Manufacturing/IoT — machine drift

Telemetry floods (high volume) → **rules-first routing** keeps LLM cost near zero; only ambiguous patterns go to the model; Field Service Case auto-created; line-stop actions HIGH-risk. *Heuristics:* model routing, semantic cache, continuous batching analogue (bounded queue), (roadmap) distillation of triage into a small model.

7.4 Healthcare — billing denials with PHI

Redaction (extended with NER) + least-privilege + audit are the *entry ticket*; self-hosted open models (quantization, vLLM serving) may be required for data residency — that is when heuristics 9–16 stop being N/A.

8 · Closing the gaps — prioritised roadmap by heuristic

Priority	Gap	Heuristic(s)	Why first	Effort
P0	Durable approvals/outbox, idempotent audit upsert	— (production hygiene)	Unblocks multi-replica and restarts	M
P0	Golden-set evals + shadow mode + CI gate	#20 Evals	Without it, no safe expansion of autonomy	M
P1	Real embeddings + hybrid search + reranker	#2, #3, #1	Biggest quality lever for grounding	M
P1	Prompt cache + semantic cache + confidence routing	#4, #15, #14	Cuts cost/latency on repetitive alerts	M
P1	MCP server exposing the registry; NER PII (Presidio); guardrails	MCP, Security	Ecosystem interoperability and defence in depth	M
P2	Langfuse/Phoenix traces; Ragas-style retrieval metrics	Observability	Debuggable agent traces	S
P2	Override-derived dataset → SFT/LoRA → distilled triage model	#5, #6, #8	Lower unit cost at volume	L
P3	GRPO with verifiable rewards	#7	Advanced; only if volume justifies	L
P3	Self-hosted serving (quantization, KV/paged attention, speculative decoding, disaggregation)	#9–12, #16	Only for sovereignty-driven customers	L



9 · Interview talking points

60-second version: *"The ecosystem has dozens of tools; the hard part is the connective tissue. My bridge is a governed action plane: it grounds the model, constrains it to typed tools, forces consequential actions through a human, and records evidence — so a regulated customer can approve it. In heuristic terms, it implements tool use and structured output, partially implements RAG, routing, hybrid retrieval and evals, and has explicit seams for caching, reranking, fine-tuning and distillation."*

If the panel asks…	Say…
"Which of the 20 heuristics did you use?"	"Two fully, five partially — and I can name the pressure that would trigger each of the rest."
"Why no fine-tuning?"	"No labels yet. Human overrides will create them; SFT/LoRA/distillation come after evals prove a gap."
"Why not quantization / vLLM?"	"Hosted models. Those become relevant when data residency forces self-hosting — I've mapped that path."
"Where does MCP fit?"	"The action manifest is already tool-shaped; exposing it as an MCP server is a thin adapter. Salesforce also offers hosted MCP servers."
"Which industry first?"	"Asset management and payments — high regulation, high volume, and where I have shipped before."
"What did you *not* build?"	"Real Kafka, durable state, Data Cloud itself, real embeddings. All listed with a priority in section 8."

> **Caution**
>
> Never describe the simulated pieces (Data Cloud grounding, hashing embeddings, offline benchmark) as production-proven. The candour is part of the pitch.